Day 7 — Sāyana vs Nirayana

Module: Movement of the Sun · **Band:** Middle · **Time:** 45 min

Learning objective

By the end of this lesson, students will distinguish the **sāyana** (tropical) and **nirayana** (sidereal) zodiacs, explain that the gap between them — the **ayanāṁśa** — is ~24° today, and link this gap to **precession** (the slow wobble of Earth's axis).

Materials

- Whiteboard
- Two transparent circles printed on overhead-projector film or paper-cutouts: each shows the 12 *rāśis* arranged around a circle
- A long printed strip showing the year (Jan to Dec) with the two solstice markers
- Student notebooks

Vocabulary introduced today

- *sāyana* (IAST: *sāyana*; lit. "with-ayana / tropical") — the reference frame fixed to the equinoxes. The Sun enters *Meṣa* (Aries) on the **March equinox** by definition. The seasons stay locked to this system.
- *nirayana* (IAST: *nir-ayana*; lit. "without-ayana / sidereal") — the reference frame fixed to the **fixed stars**. The Sun enters *Meṣa* when it physically passes a fixed point against the background stars.
- *ayanāṁśa* (IAST: *ayana-aṁśa*; lit. "precessional portion") — the angular offset between the two systems, ~24° today.
- **Precession** — the slow wobble of Earth's rotation axis. It traces a cone in space over ~25,800 years. Because of this wobble, the position of the equinoxes drifts among the stars at ~50 seconds of arc per year (~1° per 72 years).

Lesson flow

Warm-up (5 min)

- Recall the puzzle: *Makara Saṁkrānti* is ~14 January but the winter solstice is ~21 December — a 24-day gap.
- Prompt: *"24 days. Why 24?"*
- Take 2 guesses. Park them. *"Today we resolve it."*

Core (25 min)

1. **Two ways to define the start of the zodiac.**

   Imagine the great circle of the ecliptic divided into 12 *rāśis*. **Where does *Meṣa* (the first *rāśi*) begin?**

- **Option A (*sāyana* / tropical):** *Meṣa* begins at **the March equinox point**. This makes the Sun's position lock-step with the seasons. When the Sun is at 0° *Meṣa*, it's the March equinox — always. Modern Western astronomy uses this.
- **Option B (*nirayana* / sidereal):** *Meṣa* begins at **a fixed point against the stars** (chosen near the star Spica / *Citrā* by Indian convention). When the Sun is at 0° *Meṣa*, it's physically passing that star-anchored boundary. Indian traditional *jyotiṣa* uses this.

2. **Why do they differ?** Because of **precession**.

    - Earth's rotation axis isn't perfectly fixed in space. It's a slow wobble — like a spinning top tilting in a circle.
    - One full wobble takes about **25,800 years**.
    - The equinox point (where the Sun crosses the equator going north) is defined by the axis — so as the axis wobbles, the equinox point **drifts** against the background stars at about **50 seconds of arc per year**, or **1° per 72 years**.
    - The drift goes *backwards* through the zodiac — from *Meṣa* into *Mīna*, then into *Kumbha*, and so on.

3. **The size of the gap right now.**

    - The two systems were aligned approximately **1700 years ago** (in the early centuries CE).
    - In that time, precession has drifted the equinox by **~24°** against the stars.
    - So the *sāyana* zodiac and the *nirayana* zodiac are now **24° apart**.
    - **24° / 360° × 365.25 days ≈ 24.3 days**. That's the gap.
    - **24-day gap = 24° offset = ~1700 years of precession.** That's why *Makara Saṁkrānti* is 24 days *after* the winter solstice.

4. **Demo with the two transparencies.**

    - Place the two *rāśi* circles on top of each other, both starting at the top.
    - Slide one circle 24° clockwise. Now both circles still have *Meṣa, Vṛṣabha, …* in order, but the boundaries are offset.
    - "If the Sun is at 0° *Meṣa* in the bottom (sāyana) circle on March 21, where is it in the top (nirayana) circle?" — answer: it's at 0° *Mīna* + 6° (since it's offset 24° backward). It enters *Meṣa* nirayana ~24 days later, on ~April 14.

5. **A summary table:**

Event	*Sāyana* date	*Nirayana* date	Gap
Sun enters *Makara* / Capricorn	~21 December (= winter solstice)	~14 January (= Makara Saṁkrānti)	~24 days
Sun enters *Karka* / Cancer	~21 June (= summer solstice)	~16 July	~25 days
Sun enters	~21 March (= March equinox)	~14 April	~24

Event	*Sāyana* date	*Nirayana* date	Gap
Meṣa / Aries			days
Sun enters *Tulā* / Libra	~23 September (= September equinox)	~17 October	~24 days

6. **Which is "right"?** Neither — they're answering different questions.
    - *Sāyana* answers: "Where is the Sun relative to the seasons (equinoxes / solstices)?" — used by modern astronomy and the civil calendar.
    - *Nirayana* answers: "Where is the Sun against the actual background stars?" — used by traditional Indian almanacs.
    - **Both are internally consistent.** They diverge because the equinox drifts; whichever frame you fix, the other one moves.

Activity (10 min) — Compare-contrast worksheet

(See `activities/activity-03-sayana-nirayana.md` for the full sheet.)

In pairs, students fill a two-column table:

Question	*Sāyana* answer	*Nirayana* answer
Sun in *Meṣa* on …	~21 March	~14 April
Aligned with seasons?	Yes (by definition)	Drifts over centuries
Aligned with stars?	Drifts over centuries	Yes (by definition)
Used by modern astronomy?	Yes	No
Used by Indian *pañcāṅga*?	No	Yes
What changes over millennia?	Star background drifts	Season alignment drifts

Wrap-up (5 min)

- Exit ticket: *"Why is Makara Saṁkrānti on January 14 and not December 21? Answer in one sentence."*
- Acceptable answer: "Because the *nirayana* (sidereal) zodiac, used for *Makara Saṁkrānti*, is offset from the *sāyana* (tropical) zodiac, used for the solstice, by ~24° due to precession over ~1700 years."
- Preview Day 8: *"Tomorrow — build the project. Your rāśi-wheel will show BOTH systems."*

Student-facing content

> **Two zodiacs, one sky.**

System	Anchored to	Used by
Sāyana (tropical)	The **equinoxes**	Modern astronomy, Western horoscopes
Nirayana (sidereal)	The **fixed stars**	Indian traditional almanacs

The two were aligned ~1700 years ago. Earth's slow axial wobble — **precession** — has drifted them apart by **~24°** since. That gap is called the **ayanāṁśa**.

24° ÷ 360° × 365.25 days ≈ **24 days**. That's why *Makara Saṁkrānti* (Jan 14, *nirayana*) is 24 days after the winter solstice (Dec 21, *sāyana*).

Homework

- Look up the *ayanāṁśa* value for today on any astronomy website or *pañcāṅga* app. Most sources will quote either the *Lahiri* value (the Government of India standard) or the *Krishnamurti* value. They differ by a few minutes of arc. Note both.
- Write 4 sentences: explain to a younger sibling why their newspaper horoscope sign (Western, *sāyana*) might be one *rāśi* "off" from a traditional *pañcāṅga* (Indian, *nirayana*).

Differentiation

- **One tier up:** When will the *sāyana* and *nirayana* zodiacs realign? Hint: when *ayanāṁśa* completes another ~336° (= 25,800 yrs × 336/360). Approximately when? (Answer: in ~24,000 years.) What was the *ayanāṁśa* in 1 CE? (Answer: very close to 0°.)
- **One tier down:** Just remember: ~24 days difference. *Sāyana* = with the seasons. *Nirayana* = with the stars. Don't worry about precession mechanics yet.

Teacher notes

- This is the densest day of the module. Plan to take questions — and to defer some to next class. The full theory of precession is the Day 8 *senior* topic; here we just name it.
- Some students will be upset that "the calendar is wrong." Reframe: neither calendar is wrong. Each is right *within its frame*. The drift is a real astronomical phenomenon.
- If you have only 35 minutes, drop the worksheet and use the time for questions and the demo.

Citation(s) used in this lesson

- Precession rate (~50″/year, ~25,800-year period) is a modern astronomy standard value (see `sources.md`). Lahiri *ayanāṁśa* is the Government of India standard adopted in the 1950s by the Calendar Reform Committee chaired by Meghnad Saha.